



FROM BITS TO QUBITS: HOW THEY COMPARE

	Classical Computer	Supercomputer	Quantum Computer
 Tech	Uses binary bits that represent either 0 or 1	Uses thousands to millions of classical processors working in parallel at extremely high speeds	Uses quantum bits (qubits) that can exist as both 0 and 1 simultaneously through superposition
 Uses	Everyday tasks like web browsing, gaming, office work, streaming, and communication	Weather forecasting, scientific simulations, space research, AI model training, nuclear research, and complex industrial calculations	Solving highly complex optimisation, cryptography, molecular simulation, quantum chemistry, and advanced AI problems
 Strengths	Highly stable, reliable, affordable, and easy to use	Massive computational power capable of processing enormous datasets rapidly	Can theoretically solve certain problems exponentially faster than classical systems
 Limitations	Runs operations sequentially. Highly complex mathematical or cryptographic problems can take thousands of years	Extremely expensive, consumes enormous power, and still limited by classical computing architecture	Highly experimental, error-prone, requires ultra-cold environments, and currently unsuitable for everyday computing tasks
Typical Price	₹20,000 to ₹3lakh+ (depending on specifications)	₹100 crore upwards	\$10–50 billion+, 1,000 qubits

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ers with 50–1,000 physical qubits, satellite-based secure quantum communications over 2,000 kms, inter-city quantum key distribution networks, quantum sensors, atomic clocks, and indigenous quantum devices.

To drive the programme, the government has established four Thematic Hubs hosted at Indian Institute of Science, Indian Institute of Technology Madras, Indian Institute of Technology Bombay, and Indian Institute of Technology Delhi, bringing together 152 researchers across 43 institutions.

Anil Prabhakar, Principal Investigator at IIT Madras, says “there is a strong collaborative engagement between the four hubs and the technical institutes, research labs and universities that work with the hubs in a spoke-and-spike model.” IIT Madras hosts the hub for quantum communication and leads technical groups on quantum algorithms, sensing, and error correction.

THE REALITY CHECK

The global race is moving at an extraordinary speed. The US is relying heavily on private sector giants such as IBM, Google, and Microsoft to push quantum hardware and algorithms while China has invested aggressively in state-backed infrastructure and secure communications systems. Europe, meanwhile, is focusing on building long-term collaborative research ecosystems.

India is still in the capability-building phase. “Its position is best described as transitioning from exploratory research to structured capability building rather than operating at scale with commercial or near-commercial quantum systems,” says Sushovan Mukhopadhyay, Director Analyst at Gartner.

The gap is most visible in hardware. “The NQM targets the development of intermediate-scale quantum computers with 20–50 physical qubits in three years,